

A Safety-Aware Hybrid Framework for Visual Place Recognition

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Abstract

Visual Place Recognition (VPR) [6] [11] [13] constitutes a fundamental pillar for autonomous navigation and mobile robotics, providing a geographic position estimation based exclusively on visual content. However, the current state of the art is characterized by a critical dualism: efficient global descriptors that are vulnerable to perceptual aliasing, and precise local matching algorithms that are computationally burdensome. This work introduces a hybrid "Safety-Aware" framework that integrates the efficiency of global retrieval with the robustness of conditional geometric refinement. We propose an Adaptive Re-ranking mechanism based on uncertainty estimation, capable of modulating computational effort through an analysis of the geometric ambiguity ratio. Our contributions include an extensive evaluation of models based on CNN and MLP-Mixing (MixVPR, CosPlace) and the implementation of a safety pipeline that guarantees certified predictions in mission-critical scenarios, distinguishing between correct matches and "Uncertain Location" states.

1. Introduction

Determining geographic position based exclusively on visual data, known as Visual Place Recognition (VPR), is a task of fundamental importance for autonomous navigation, augmented reality, and mobile robotics. VPR is essential where traditional systems, such as GPS, fail systematically due to signal obscuration or reflection, typically in indoor environments, tunnels, or so-called "urban canyons". Formally, VPR is treated as a large-scale image retrieval problem [11][6]. Given a massive database of "geo-tagged" images (associated with GPS coordinates and orientation), the goal is to identify the corresponding place in the map for a given "Query" image. The scientific challenge lies in find-

ing a representation that is invariant to drastic variations in visual appearance:

- **Illumination and Atmospheric Conditions:** Recognizing a place in night conditions by comparing it with a daytime database.
- **Seasonality:** Managing radical changes in vegetation and landscape (e.g. the presence of snow vs. summer foliage).
- **Structural Changes and Dynamism:** Ignoring transient elements like vehicles, pedestrians, or construction sites, focusing on stable architectural structures.

Although the use of Deep Convolutional Neural Networks (DCNN) and modern architectures has improved the robustness of semantic representations, current deterministic models suffer from overconfidence, which is the system's tendency to provide a unique prediction with a high degree of certainty even in the absence of sufficient visual evidence or in situations of blatant ambiguity—an extremely dangerous behavior in safety-critical contexts. In this work, we highlight the need for integrated uncertainty estimation, understood as the framework's algorithmic capacity to self-evaluate its own reliability to signal doubtful cases instead of producing potentially erroneous localizations. We introduce a methodological extension that quantifies localization solidity via the counting of geometric inliers [15]. These constitute the set of local correspondences that strictly respect the constraints of a coherent spatial transformation (such as homography), ensuring that the match is supported by a shared physical structure and not by random visual similarity. This approach allows the system to signal a state of uncertainty when visual evidence is lacking or the match is geometrically ambiguous [11].

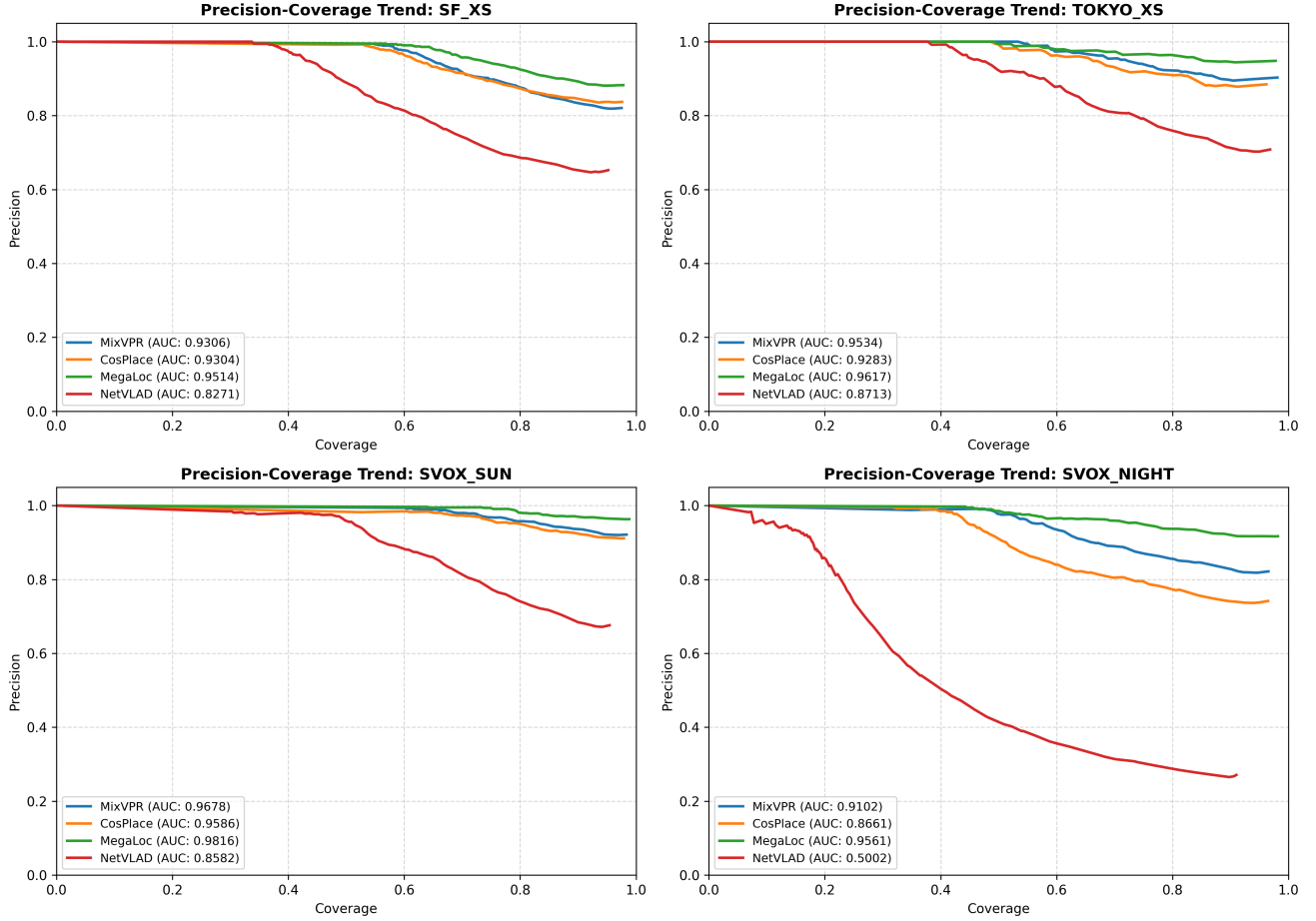


Figure 1. Quantitative comparison of the proposed framework against state-of-the-art methods. We report the AUC (Area Under Curve) on four challenging datasets.

2. Related work

The literature on Visual Place Recognition (VPR) has undergone a radical transformation in the last decade, evolving from manual heuristics to end-to-end deep learning architectures [5] [6] [11] [18].

From Hand-crafted Methods to Semantic Representations Before the advent of Convolutional Neural Networks (CNNs), VPR relied on hand-crafted local descriptors, such as SIFT or SURF, designed to extract points of interest (corners or edges). These were aggregated via frameworks like Bag-of-Words (BoW) or VLAD (Vector of Locally Aggregated Descriptors), which quantized features into a visual vocabulary [3] [6]. However, relying on raw pixel gradients, these methods failed systematically in the presence of illumination changes (day/night) or seasonal change [7]. The CNN revolution allowed to learn high-level semantic features, condensing them into Global Descriptors

(embeddings) capable of recognizing the structure of buildings and skylines while ignoring superficial variations like shadows or weather conditions [13] [7].

2.1. Global Representation Learning

Modern global aggregation methods aim to generate compact and discriminative embeddings:

- **NetVLAD:** The work of Arandjelovic et al.(2016) [3] introduced NetVLAD, the first differentiable aggregation layer for end-to-end training. Its innovation lies in Soft-Assignment, which replaces the Hard Assignment (where every feature belongs to only one cluster) of classic VLAD with a Softmax function to probabilistically assign a feature to all clusters. For training, it uses Triplet Ranking Loss on weakly supervised data (Google Street View Time Machine), teaching the network to pull the Query closer to a positive image (same place) and push it away from a negative one (different place).

- **CosPlace:** Berton et al. (2022) [5] criticized the limitation of Contrastive Learning used by NetVLAD, which requires an extremely slow negative mining process (searching for different but similar images) on databases. It redefined VPR as a classification task by splitting the earth’s surface into discrete geographic cells (Class IDs), it eliminates the need for expensive mining of negatives; typical of Contrastive Learning. CosPlace learns extremely compact (512-D) and scalable descriptors, reducing GPU memory usage.
- **MixVPR:** Introduces an architectural shift based on MLP (Multi-Layer Perceptrons) [2]. Instead of aggregating locally, MixVPR “mixes” feature maps globally. This approach allows every neuron to have a global receptive field, capturing long-range relationships in the image in an extremely lightweight and high-performance manner.
- **MegaLoc:** Addresses domain fragmentation by proposing a universal model capable of excelling simultaneously in urban scenarios, landmark retrieval, and indoor environments [4]. The innovation lies in the use of multi-similarity loss and balanced sampling on heterogeneous datasets, ensuring a robust and generalizable visual representation.

2.2. Training paradigm and metric learning

To understand how these models learn, one must distinguish between:

- **Contrastive Learning (e.g. NetVLAD/MixVPR):** Relies on Triplet Ranking Loss to minimize the distance between the Query and a Positive (same place) and maximize it with a Negative (different place) [3] [2].
- **Classification (e.g. CosPlace):** Treats every geographic cell as a discrete class. The network learns to recognize fixed features of the cell, making training much faster and scalable on massive datasets like GSV-Cities [5][1].

2.3. Local Feature Matching and Geometric Re-ranking

To mitigate perceptual aliasing (different places that are visually similar), the standard pipeline introduces a Refinement or Re-ranking phase.

- **SuperPoint:** A fully convolutional network that simultaneously detects keypoints and local descriptors in a single pass. Self-supervised training via Homographic Adaptation guarantees robustness to perspective changes [8].
- **SuperGlue:** A matcher based on Graph Neural Networks (GNN) that uses Self and Cross-Attention mechanisms to solve matching as an optimal transport problem via the Sinkhorn algorithm [14].
- **LightGlue:** Drastically optimizes the SuperGlue process by introducing adaptivity. Thanks to the Early Stopping mechanism, the network interrupts calculation on “easy”

image pairs, ensuring speed without loss of precision [10].

- **LoFTR:** A detector-free matcher ideal for contexts with poor texture (uniform walls, skies). It operates on dense feature maps via Transformers with a coarse-to-fine approach. Finally, spatial verification relies on the RANSAC (Random Sample Consensus) paradigm [9]. RANSAC operates via an iterative sampling cycle to estimate a coherent geometric transformation (homography). This process separates Inliers (matches that respect the geometry) from Outliers (matching errors). The final count of Inliers, termed the Consensus Set, provides the statistical confidence metric necessary to certify the localization and forms the basis for our extension on uncertainty management [16] [18].

3. Methodology

We propose a hierarchical two-stage Visual Place Recognition (VPR) framework designed to balance large-scale efficiency with geometric reliability. The pipeline combines global image retrieval with adaptive geometric verification, augmented by a **safety-aware uncertainty estimation module** to mitigate perceptual aliasing and model overconfidence.

3.1. Stage 1: Global Retrieval

The first stage reduces the search space by retrieving a shortlist of K candidate locations (we set $K = 20$) from a large-scale geo-referenced database.

Offline Database Indexing Each database image is encoded using a shared backbone and aggregation architecture (e.g., MixVPR or CosPlace), producing a compact global descriptor. All descriptors are stored in a vector index (FAISS), enabling sub-linear nearest-neighbor search in high-dimensional spaces. This pre-processing is performed once, allowing the original images to be archived and minimizing runtime memory usage.

Online Query Processing

Given a query image, a global descriptor is computed using the same network weights. The descriptor is compared against the indexed database using K -NN search with either Euclidean distance or dot-product similarity (equivalent to cosine similarity for normalized vectors).

Output: Stage 1 returns an ordered shortlist $\mathcal{L} = \{I_1, \dots, I_K\}$ of candidate images and their associated UTM coordinates. This set defines the hypotheses evaluated during geometric refinement.

3.2. Stage 2: Geometric Re-ranking

Global retrieval relies solely on appearance similarity and is therefore vulnerable to perceptual aliasing. To address this limitation, Stage 2 performs geometric re-ranking by

Table 1. VPR Performance. Results are reported as **Recall@1 / 5 / 10 / 20**. Bold indicates the best retrieval performance for each backbone.

Method	Tokyo_XS	SF_XS	SVOX_SUN	SVOX_NIGHT
	R@1 / 5 / 10 / 20	R@1 / 5 / 10 / 20	R@1 / 5 / 10 / 20	R@1 / 5 / 10 / 20
Megaloc (DINOv2)	99.0	91.5	99.6	99.3
+ SuperPoint + LG	94.3 / 98.4 / 98.7 / 99.0	87.0 / 90.6 / 91.3 / 91.5	95.8 / 98.9 / 99.4 / 99.6	90.9 / 97.4 / 98.7 / 99.3
+ SuperGlue	93.0 / 98.1 / 98.4 / 99.0	85.8 / 89.9 / 90.7 / 91.5	94.1 / 98.5 / 99.3 / 99.6	90.4 / 97.8 / 98.5 / 99.3
+ LoFTR	94.3 / 97.8 / 98.7 / 99.0	86.5 / 89.7 / 90.8 / 91.5	97.1 / 99.3 / 99.5 / 99.6	93.1 / 98.7 / 99.0 / 99.3
CosPlace (ResNet50)	91.4	86.9	95.0	78.6
+ SuperPoint + LG	86.0 / 89.2 / 89.8 / 91.4	82.5 / 85.6 / 86.3 / 86.9	90.2 / 93.6 / 94.6 / 95.0	71.8 / 75.8 / 76.9 / 78.6
+ SuperGlue	84.8 / 88.6 / 90.2 / 91.4	81.5 / 84.9 / 86.3 / 86.9	86.7 / 93.0 / 94.1 / 95.0	71.3 / 75.9 / 77.5 / 78.6
+ LoFTR	86.7 / 89.2 / 90.5 / 91.4	81.7 / 84.7 / 86.1 / 86.9	92.3 / 93.9 / 94.4 / 95.0	72.9 / 76.9 / 77.9 / 78.6
NetVLAD (VGG16)	74.6	65.3	69.0	29.5
+ SuperPoint + LG	68.9 / 71.7 / 73.0 / 74.6	62.5 / 64.7 / 65.0 / 65.3	65.0 / 66.5 / 67.6 / 69.0	25.0 / 27.5 / 28.4 / 29.5
+ SuperGlue	67.0 / 71.1 / 72.7 / 74.6	60.6 / 64.1 / 64.6 / 65.3	64.2 / 66.4 / 67.8 / 69.0	25.0 / 27.3 / 28.7 / 29.5
+ LoFTR	68.9 / 70.8 / 73.0 / 74.6	61.5 / 63.9 / 64.4 / 65.3	64.8 / 66.6 / 67.7 / 69.0	25.5 / 27.6 / 28.9 / 29.5
MixVPR (ResNet50)	94.3	83.6	95.4	86.8
+ SuperPoint + LG	89.2 / 92.1 / 93.7 / 94.3	80.5 / 82.5 / 83.2 / 83.6	91.2 / 94.3 / 95.1 / 95.4	80.7 / 85.0 / 86.0 / 86.8
+ SuperGlue	88.9 / 92.4 / 93.3 / 94.3	79.3 / 82.2 / 83.2 / 83.6	89.7 / 93.7 / 95.1 / 95.4	79.2 / 84.6 / 86.3 / 86.8
+ LoFTR	90.2 / 93.0 / 93.7 / 94.3	79.4 / 82.8 / 83.5 / 83.6	93.1 / 95.0 / 95.2 / 95.4	81.1 / 85.4 / 86.3 / 86.8

Table 2. Comparison of Visual Place Recognition methods across datasets. The parameter τ_{adaptive} is fixed at 70 for all experiments, calibrated on the validation set of SF-XS. The table reports the specific **Ambiguity Threshold** (τ_{amb}) used to determine the **Coverage @ 99% Safety** (the percentage of queries correctly localized within 25m tolerance while maintaining $\geq 99\%$ precision).

Dataset	MixVPR (ResNet-50)		CosPlace (ResNet-50)		MegaLoc (DINOv2)		NetVLAD (VGG-16)	
	Cov.	τ_{amb}	Cov.	τ_{amb}	Cov.	τ_{amb}	Cov.	τ_{amb}
Tokyo-XS	58.41%	0.55	49.52%	0.46	56.83%	0.64	38.41%	0.34
SVOX-Sun	61.71%	0.44	60.54%	0.36	75.41%	0.62	30.21%	0.09
SVOX-Night	41.75%	0.25	36.57%	0.29	48.72%	0.52	7.29%	0.11
SF-XS	57.30%	0.40	53.20%	0.41	60.10%	0.45	36.70%	0.30

verifying spatial consistency between the query and each candidate.

Local Feature Matching All images are resized to 512×512 pixels to balance accuracy and computational cost. Local keypoints and descriptors are extracted using a fully convolutional network, followed by an attention-based matcher that estimates correspondences between query and candidate images [11][6].

Geometric Verification Spatial consistency is validated using RANSAC to estimate a geometric transformation. Matches that are inconsistent with the model are discarded as outliers. The number of inliers defines the geometric confidence score for each candidate. Candidates are re-ranked based on decreasing inlier count. The top-ranked image is selected as the geometric winner, and its UTM coordinates are returned as the localization estimate [9].

4. Extension

While geometric verification improves precision, deterministic pipelines remain prone to overconfidence. We introduce a safety-aware extension that explicitly quantifies geometric ambiguity and allows the system to abstain when evidence is insufficient. The proposed localization framework transitions from a coarse global search to a high-precision geometric verification, balancing computational efficiency with operational safety. In the initial phase, the system performs global retrieval to process a vast database of images and extract a restricted shortlist of the 20 most likely candidates for each query. This process utilizes state-of-the-art deep neural networks, including **NetVLAD**, **MixVPR**, **CosPlace**, and **MegaLoc**, to generate compact global descriptors. These descriptors condense the semantic information of the scene, such as buildings and road structures, into numerical vectors. Similarity is then calculated using L2 distance or Dot Product metrics to prioritize speed over millimetric precision. [3][2][4][5][6]

4.1. Adaptive Efficiency and Re-ranking

To optimize the pipeline for real-time applications, we implement an **Adaptive Efficiency Check**. For each query, the system first evaluates the geometric consistency of the top-ranked candidate from the retrieval phase. We define n_1 as the count of geometric inliers—keypoints that pass a consistency check via algorithms like RANSAC [9].

- **Early Exit:** If $n_1 > \tau_{\text{adaptive}}$, where τ_{adaptive} is a threshold calibrated on a validation set to guarantee $\geq 99\%$ localization precision, the prediction is immediately accepted. This mechanism avoids the significant computational cost of matching the remaining candidates in the shortlist when the first result is already highly reliable [15].
- **Conditional Re-ranking:** If the top candidate fails this check ($n_1 \leq \tau_{\text{adaptive}}$), the system proceeds to a full geometric re-ranking of all K candidates in the shortlist. During this phase, local features are extracted using SuperPoint and matched via LightGlue (or SuperGlue) to identify the true location. The candidate with the highest number of inliers is then promoted to the first position (r_1), regardless of its initial vector similarity score.

4.2. Geometric Ambiguity Detection

Even after re-ranking, environmental challenges such as perceptual aliasing, where different locations appear visually identical—can lead to false-positive results. To mitigate this, we introduce a **safety gate** based on the **geometric ambiguity ratio** ρ . We define this ratio as follows:

$$\rho = \frac{n_1}{n_2}, \quad \rho \in [0, 1] \quad (1)$$

where n_1 and n_2 denote the inlier counts of the best and second-best candidates in the re-ordered shortlist, respectively. A localization prediction is accepted only if it satisfies the condition:

$$\rho < \tau_{\text{amb}} \quad (2)$$

where τ_{amb} represents a predefined safety threshold. If the ratio exceeds this limit, it indicates that the system cannot confidently distinguish between the top two candidates. In such cases, the system outputs an *Uncertain* state. This preventatively blocks false-positive localizations and enables the vehicle to trigger fallback strategies, such as sensor fusion with LiDAR or IMU data.

4.3. Safety-Coverage Trade-off

To evaluate the robustness of this safety gate, we perform a **Safety-Coverage** analysis. By sweeping the threshold τ_{amb} across the range $[0, 1]$, we generate a curve that illustrates the trade-off between the percentage of queries

the system attempts to localize (coverage) and the resulting precision (safety). We measure the Area Under the Curve (AUC) using the trapezoidal rule. This threshold-independent metric captures the intrinsic ability of the combined **Cosplace**, **MixVPR**, **NETVLAD** and **MegaLOC** retrieval and geometric matching pipeline to effectively discriminate between valid localizations and geometric ambiguities.[2][3][5]

5. Experimental Results and Ablation studies

We evaluate the proposed framework using a systematic experimental protocol designed to isolate the contribution of each pipeline component and analyze the trade-off between localization accuracy, computational efficiency, and safety. All experiments follow a controlled ablation strategy inspired by standard VPR benchmarking practices.[6][11][12]

5.1. Experimental Setup and Datasets

We conduct experiments on four datasets selected for their complementary characteristics and increasing levels of difficulty:

- **GSV-XS:** a reduced version of GSV-Cities, used exclusively for training. Unlike weakly supervised datasets, GSV-XS provides accurate ground truth with camera orientation (bearing), enabling fully supervised learning and stable feature convergence [1].
- **SF-XS:** derived from SF-XL, split into a validation set (SF-XS Val) for threshold calibration and a test set (SF-XS Test) for evaluation in dense urban environments [5].
- **Tokyo-XS:** a subset of Tokyo 24/7, serving as the reference benchmark for extreme appearance changes caused by day-night transitions and viewpoint variations [17].
- **SVOX:** a cross-domain test set used to evaluate generalization across different cities, weather conditions, and illumination regimes. In our experiments, we followed the standard SVOX-Sun and SVOX-Night division to specifically assess model performance under daytime sunny conditions (SVOX-Sun) and nighttime low-light scenarios (SVOX-Night), enabling robust evaluation of illumination invariance. [6].

5.2. Evaluation Metrics

Visual Place Recognition is traditionally evaluated using Recall@N, defined as the percentage of queries for which at least one of the top- N retrieved images lies within a geographic tolerance (25 m) from the ground truth [11] [6].

However, Recall@N does not capture prediction reliability. For safety-critical applications, we therefore introduce the following complementary metrics:

- **Safety Rate (S):** the proportion of geographically correct predictions among all accepted (non-uncertain) queries.

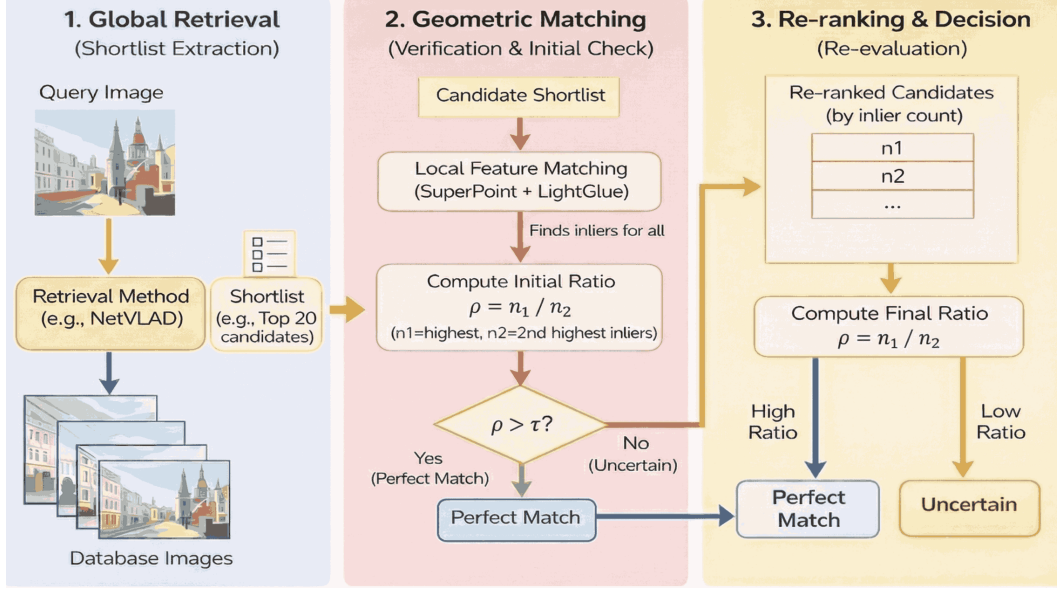


Figure 2. Visual pipeline of the extension

- **Coverage (C):** the fraction of queries for which the system outputs a localization estimate.
- **AUC Safety-Coverage:** a threshold-independent metric obtained by integrating the Safety-Coverage curve over all values of the ambiguity threshold.

A high AUC indicates a strong intrinsic capability of the model to discriminate between correct localizations and geometrically ambiguous false positives.

5.3. Stage 1 Ablation: Global Retrieval

We analyze the impact of different global retrieval architectures:

- **NetVLAD:** VGG-16 backbone with 4096-D descriptors. While robust, it exhibits high memory consumption and K-NN latency [3].
- **CosPlace:** ResNet-50 backbone producing compact 512-D descriptors, offering a favorable balance between recall and efficiency [5].
- **MixVPR:** ResNet-50 truncated at Layer 3 with MLP-based feature mixing. This configuration achieves state-of-the-art efficiency with minimal loss in retrieval accuracy [2].
- **MegaLoc:** DINOv2-based foundation model generating high-dimensional (8448-D) descriptors, prioritizing semantic richness over efficiency [4].

Results highlight the importance of descriptor compactness for large-scale deployment, with MixVPR and CosPlace providing the best efficiency-accuracy trade-off while MegaLoc provides the best accuracy at the cost of increased computational overhead.

5.4. Stage 2 Ablation: Local Feature Matching and Re-ranking

We evaluate multiple geometric refinement strategies, including LoFTR, SuperGlue, and SuperPoint with LightGlue [16][14][10].

While all methods improve recall by resolving perceptual aliasing, their computational profiles differ significantly. LightGlue achieves comparable geometric precision to SuperGlue while substantially reducing latency through adaptive early stopping, making it particularly suitable for large-scale and real-time scenarios.

5.5. Uncertainty Quantification

The integration of uncertainty estimation addresses a critical limitation of deterministic VPR systems: overconfidence. By explicitly modeling geometric ambiguity, the framework prevents high-confidence false positives that would otherwise pass standard geometric verification.

Adaptive Safety Framework The efficiency threshold τ_{adaptive} is calibrated on SF-XS Val through statistical binning of queries based on the inlier count of the top retrieved candidate, with our τ_{adaptive} set to 70. We select a conservative threshold corresponding to $\geq 99\%$ geographic precision. Since this stage bypasses ambiguity checks, a conservative choice is essential to guarantee safety.

Ambiguity Threshold Analysis During evaluation, the ambiguity threshold τ_{amb} is swept from 0 to 1 with a step of 0.01. This fine-grained resolution enables both accurate

AUC computation via trapezoidal integration and precise operating-point selection. As τ_{amb} increases, coverage improves at the cost of reduced safety, reflecting the acceptance of increasingly ambiguous matches.

Comparative Analysis Our experimental results reveal a distinct performance hierarchy driven by the underlying architectural strengths of each backbone. **Megaloc (Dinov2)** emerges as the superior encoder for challenging, unstructured environments, demonstrating exceptional illumination invariance. It dominates the *SVOX Night* dataset with 48.72% coverage—a scenario where traditional methods collapse—and leads in *SVOX Sun* (75.41%) and *SF-XS* (60.10%). This suggests that the self-supervised features of Dinov2 provide the critical semantic understanding needed for extreme lighting changes. In contrast, **MixVPR (ResNet50)** proves to be the most reliable “all-rounder” for standard urban navigation, securing the top spot in *Tokyo XS* (58.41%) and consistently ranking as the runner-up. While **CosPlace** offers stable mid-tier performance, **NetVLAD** shows significant obsolescence, particularly in night scenes where its coverage drops to a negligible 7.29%, rendering it unsuitable for 24-hour autonomous operations without safety filtering.

6. Impact of the Safety-Aware Extension

The decisive advantage of our proposed framework lies in its ability to convert raw retrieval scores into actionable, safety-critical decisions. Unlike standard reranking—which merely attempts to re-order candidates but eventually forces a prediction—our *Safety-Aware extension* introduces a proactive “veto” power based on a geometric ambiguity check ($\rho = n_1/n_2$).

This distinction is visually captured in the Precision-Coverage (AUC) graphs 1. By integrating the curves starting from the origin ($x = 0$), we explicitly demonstrate that the system maintains a **constant precision of 1.0** during the initial phase of retrieval. This horizontal plateau represents the “Safe Zone”: a regime where the extension identifies high visual ambiguity and correctly *abstains* from responding rather than risk a mislocalization.

This selective capability is most evident when evaluating weaker models under environmental stress. For instance, in the challenging *SVOX Night* dataset, rather than forcing **NetVLAD** to produce erroneous coordinates, the system identifies the lack of distinct features and abstains from 93% of the queries. Conversely, the robust feature extraction of **MixVPR** allows for a significantly longer plateau of perfect precision before the first errors occur. By “sanitizing” the backbone’s output, the extension ensures that localized results achieve at least 99% precision. This validation confirms that the framework is not merely a performance optimizer, but a necessary safety layer for autonomous vehicles,

where the catastrophic cost of a false positive—being “confidently wrong”—far outweighs the benefit of a marginal increase in raw recall.

6.1. Efficiency vs. Reliability Trade-off

We compared our adaptive safety framework against a baseline default strategy that performs exhaustive geometric re-ranking on all shortlist candidates. While the default approach aims to maximize recall, it suffers from significant limitations that we addressed in our framework:

- **Computational Overhead:** Standard re-ranking is blind to query difficulty, incurring high latency even for “easy” queries that could be solved by global retrieval alone. By utilizing image retrieval models like **NetVLAD** and **MixVPR** to extract candidates, our framework applies an *Adaptive Efficiency Check*. If the top candidate meets the τ_{adaptive} threshold, an early exit is triggered, saving substantial processing time.
- **Mitigating Model Overconfidence:** Traditional systems often exhibit overconfidence, providing a single GPS coordinate even when the visual evidence is ambiguous. Our framework explicitly targets this by evaluating the geometric ambiguity ratio $\rho = n_1/n_2$. By identifying cases of perceptual aliasing where two different locations appear geometrically similar the system prevents dangerous false-positive localizations.

6.2. Certified Safety via AUC Analysis and future work

The integration of the *Uncertain* state allows the system to admit doubt in scenarios where the environment is geometrically indistinguishable. Overall, the Area Under the Curve (AUC) analysis of the Safety-Coverage metric provides a threshold-independent validation of the system’s performance. It demonstrates that the proposed framework does not merely maximize Recall@N, but optimizes the intrinsic balance between efficiency, accuracy, and safety. This makes the system uniquely suited for mission-critical deployment in real-world applications.

7. Conclusion

This research has presented a hierarchical framework for Visual Place Recognition (VPR) that effectively bridges the gap between large-scale computational efficiency and operational reliability. By addressing the inherent overconfidence of deterministic retrieval models, our proposed *Safety-Aware* extension introduces a self-assessment capability that is essential for mission-critical autonomous navigation.

Ultimately, the state-of-the-art VPR methods are typically evaluated on forced-choice retrieval (Recall@1), this metric fails to capture the risks of deployment in autonomous systems. Our work introduces a safety-aware ge-

ometric extension that allows the system to abstain from prediction in the presence of perceptual aliasing. We demonstrate that this approach converts VPR into a high-precision navigation module, achieving 99% localization safety. Furthermore, we identify Megaloc (Dinov2) as the superior backbone for illumination-invariant autonomy, maintaining 48% safe coverage in night environments where traditional baselines require complete disengagement. This work demonstrates that for autonomous systems to operate safely in dynamic urban environments, they must not only see accurately but also possess the meta-cognitive ability *to know when they do not know*. This framework provides a robust foundation for the next generation of visual localization systems, where safety and computational intelligence are deeply intertwined.

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